

Inverses from Graphs and Tables

TEACHER KEY | Algebra II Honors | Unit 1 | Day 10 | TEK A2.2.B

Objective: I can find the inverse of a relation from a table or graph, use $f^{-1}(x)$ notation correctly, justify why domain and range exchange, and predict whether an inverse will be a function.

Vocabulary

Inverse — the relation that **undoes** the original. Written $f^{-1}(x)$, which does **not** mean $1/f(x)$.

Building it — in a table or a set of pairs, **swap x and y**. On a graph, reflect across **$y = x$** .

One-to-one — no output is repeated. Test with a **horizontal** line: if it meets the graph twice, the function is not one-to-one.

Part 1 — Inverses from Pairs and Tables

a. $f = \{ (-4, -6), (0, -1), (2, 4), (3, -7), (7, 7) \}$

$f^{-1} = \{ (-6, -4), (-1, 0), (4, 2), (-7, 3), (7, 7) \}$

b. Complete the inverse table.

x	-4	0	2	3	7
f(x)	-6	-1	4	-7	7

x	-6	-1	4	-7	7
$f^{-1}(x)$	-4	0	2	3	7

c. Which point in f is unchanged by the swap, and what is special about it? **(7, 7) - it lies on $y = x$**

Part 2 — Why Domain and Range Exchange

Argue this in general, not from one example.

If (a, b) belongs to f , then **(b, a)** belongs to f^{-1} by definition of the swap.

So every input of f^{-1} was an **output** of f . Therefore:

domain of $f^{-1} =$ **range of f** range of $f^{-1} =$ **domain of f**

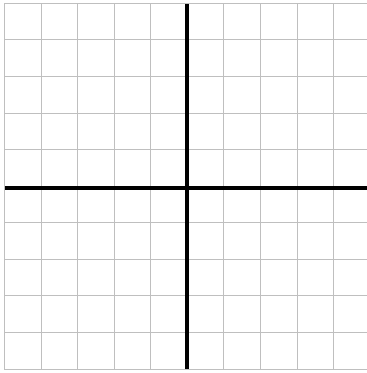
f	Domain of f	Range of f	Domain of f^{-1}	Range of f^{-1}
x^3	all reals	all reals	all reals	all reals
2^x	all reals	$(0, \infty)$	$(0, \infty)$	all reals
$\sqrt{x}$	$[0, \infty)$	$[0, \infty)$	$[0, \infty)$	$[0, \infty)$

Row 2 explains something from Day 7. What? **2^x and $\log_2$ are inverses, so their D and R swapped**

Part 3 — Graph a Pair

Each square is 1 unit. Plot f , then f^{-1} , then draw $y = x$.

$$f = \{ (-4, -2), (0, 0), (1, 1), (2, 4) \}$$



Record it.

Reflection line: $y = x$

Fixed points: $(0,0)$ and $(1,1)$

If (a,b) is on f , then on f^{-1} :

(b, a)

Is f one-to-one here? **yes**

$$f^{-1} = \{ (-2,-4), (0,0), (1,1), (4,2) \}$$

Part 4 — Will the Inverse Be a Function?

Apply the horizontal line test to the original, before building the inverse.

a. $g = \{ (2, 6), (4, 6), (5, 1) \}$. Write g^{-1} : $\{ (6,2), (6,4), (1,5) \}$

Is g^{-1} a function? **no** Why? **the input 6 has two different outputs**

What was true of g that caused it? **g repeated the output 6, so g is not one-to-one**

b. Which of these parent functions have inverses that are functions? Circle:

x^3 x^2 2^x $|x|$ $\sqrt[3]{x}$ Answer: **x^3 , 2^x , $\sqrt[3]{x}$**

What do the ones that fail have in common? **they turn around, so outputs repeat**

Part 5 — Going Further

1. Does $f^{-1}(x)$ mean $1/f(x)$? Give a specific counterexample with numbers.

For $f(x) = x + 3$: $f^{-1}(x) = x - 3$ but $1/f(x) = 1/(x + 3)$ — different functions.

2. A function is its own inverse. What must be true of its graph? **symmetric about $y = x$**

3. Can a function have two different inverses? Argue yes or no.

No — **the swap is determined, so the inverse relation is unique**

Exit Ticket

For $h = \{ (1, 4), (2, 9), (5, 0) \}$, write h^{-1} , give its domain and range, and say whether it is a function with a reason.

$h^{-1} = \{ (4,1), (9,2), (0,5) \}$ Domain $\{ 0, 4, 9 \}$ Range $\{ 1, 2, 5 \}$

A function? **yes - h was one-to-one, so no input repeats**